

7. Load Profile

7.1 Introduction

A load profile defines the number of virtual users, user arrival pattern, execution duration, and workload distribution used during performance testing. It represents how user traffic is generated and applied to the application during test execution.

In this project, Gatling was used to simulate multiple virtual users interacting with the BlazeDemo Flight Booking Application. The load profile was designed to mimic real-world user activity and evaluate the application's performance under concurrent access conditions.

The load was gradually introduced into the system using Gatling's ramp-up strategy to observe application behavior as user traffic increased.

7.2 Test Load Configuration

The following load configuration was used for all test scenarios:

Parameter	Value
Load Testing Tool	Gatling
Virtual Users	20 Users
Ramp-Up Duration	30 Seconds
Injection Model	Ramp Users
Test Type	Load Testing
User Arrival Pattern	Gradual
Execution Mode	Concurrent Users

The load profile was applied consistently across all implemented scenarios to ensure uniform performance measurement.

7.3 Virtual Users

Virtual users represent simulated users generated by Gatling to perform application transactions automatically.

In this project, a total of **20 virtual users** were configured for each test scenario.

Each virtual user performed the same sequence of actions that a real user would perform while interacting with the BlazeDemo application.

Examples of user activities include:

- Accessing the home page
- Searching available flights
- Selecting a flight
- Entering booking information
- Completing a purchase

The use of virtual users allows performance testing without requiring actual human participants.

Configuration Used

```
rampUsers(20)  
.during(Duration.ofSeconds(30))
```

This configuration gradually injects twenty users into the application over a period of thirty seconds.

7.4 Ramp-Up Duration

Ramp-up duration defines the time required to introduce all virtual users into the application.

For this project:

- Total Users = 20
- Ramp-Up Time = 30 Seconds

Instead of launching all users simultaneously, Gatling gradually introduced users over the specified period.

Purpose of Ramp-Up

The ramp-up strategy was selected because it closely resembles real-world application usage.

Benefits include:

- Prevents sudden traffic spikes
- Simulates natural user arrival patterns
- Allows observation of system behavior during increasing load
- Produces more realistic performance results

Example

At the beginning of execution:

- Few users are active

As time progresses:

- Additional users are added

After 30 seconds:

- All 20 users are active

This gradual increase helps evaluate how the application responds as user traffic grows.

7.5 Concurrent User Simulation

Concurrent users are users who interact with the application at the same time.

One of the primary objectives of load testing is to determine whether the application can process multiple user requests simultaneously without performance degradation.

In this project, Gatling simulated concurrent users performing the following activities:

Scenario 1

Multiple users searched for flights simultaneously.

Scenario 2

Multiple users searched for flights and selected available flights simultaneously.

Scenario 3

Multiple users completed the entire booking workflow simultaneously.

The concurrent execution of these activities helped assess:

- Application responsiveness
- Request processing capability
- Server stability
- Transaction success rate

The BlazeDemo application successfully handled all simulated concurrent user requests without failures.

7.6 Request Distribution

Request distribution refers to the manner in which requests are generated and sent to the application during test execution.

Each scenario generated different numbers of requests depending on the complexity of the workflow.

Scenario 1 – Search Flights

Requests Generated:

- Open Home Page
- Find Flights

Total workflow focused on flight search functionality.

Scenario 2 – Choose Flight

Requests Generated:

- Open Home Page
- Find Flights
- Choose Flight

Additional requests were generated due to the flight selection process.

Scenario 3 – Complete Booking

Requests Generated:

- Open Home Page
- Find Flights
- Choose Flight
- Purchase Flight

This scenario generated the highest number of requests because it included the complete booking transaction.

As the complexity of the scenario increased, the number of requests and processing activities also increased.

7.7 Load Distribution Across Scenarios

The same user load configuration was applied to all scenarios.

Scenario	Virtual Users	Ramp-Up Time
Search Flights	20	30 Seconds
Choose Flight	20	30 Seconds
Complete Booking	20	30 Seconds

Maintaining a consistent load profile allowed accurate comparison of performance metrics between scenarios.

7.8 Load Profile Summary

The load profile for this project consisted of twenty virtual users gradually introduced over a thirty-second period using Gatling's ramp-up model. The configuration was designed to simulate realistic user behavior and evaluate the BlazeDemo application's ability to handle concurrent traffic.

The selected load profile successfully generated user activity across all implemented scenarios and provided reliable performance metrics for response time analysis, throughput measurement, and system stability evaluation.